Assessing theory of mind in children: a tablet-based adaptation of a classic picture sequencing task

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Abstract: Correctly assessing children's theory of mind (TOM) is essential in clinical practice. Yet, most tasks heavily rely on language, which constitutes an obstacle for several populations. Langdon and Coltheart's (1999) Picture Sequencing Task (PST), developed for research, avoids this limitation through a minimally-verbal procedure. We thus developed a tablet adaptation of this task for individual application, engaging children's motivation and allowing response times collection. To assess this tablet-PST, we first tested a large sample of typically developing children (6-11 years-old, N = 248), which confirmed the task's structural and content validity, and permitted the construction of three standardized clinical indices. In a second experiment, we applied those to previously diagnosed autistic children (N = 23), who were expected to show atypical TOM performances. Children's outcomes were consistent with what was hypothesized and confirmed the task's clinical sensitivity and external validity. The tablet-PST thus appears as a suitable tool, providing detailed profiles to inform clinical decisions.

Keywords: theory of mind; clinical assessment; psychological tests; psychometrics; autism spectrum disorder; neurodevelopmental disorders [7]

Theory of mind (TOM) refers to the ability to attribute mental states in order to explain and predict people's behaviors (Premack & Woodruff, 1978). TOM is thought to be essential for human social interactions, grounding communication, self-representation, and moral reasoning (Waytz et al., 2010). Although TOM emerges in the first months of development (Frith & Frith, 2003), studies have found that TOM skills keep maturing up to late adolescence (e.g. Bosco et al., 2014). Critically, TOM abilities are also characterized by interindividual differences that relate to children's social and cognitive skills. For example, early TOM skills can predict later social preference, friendship quality or academic achievement (Blair & Razza, 2007; Dunn et al., 2002; Fink et al., 2014). TOM impairments also affect quality of life in several clinical populations, such as schizophrenic patients (Maat et al., 2012), individuals with traumatic brain injuries (Bivona et al., 2015), children with developmental language disorders (Botting & Conti-Ramsden, 2008) and children with autism spectrum disorder (from now on: *autism*). In clinical practice, it is therefore of obvious importance to correctly assess TOM skills in order to detect a patient's difficulties and guide interventions.

Autism is exemplary of a case that involves TOM difficulties, as these impairments have been postulated to be central or primary to this condition (Baron-Cohen, 1997). This led to a large body of research that eventually rephrased it in terms of *differences* instead of a *deficit,* and that questioned both the primacy and the universality of this marker while confirming the overall association between autism and TOM divergences (see Fletcher-Watson & Happé,

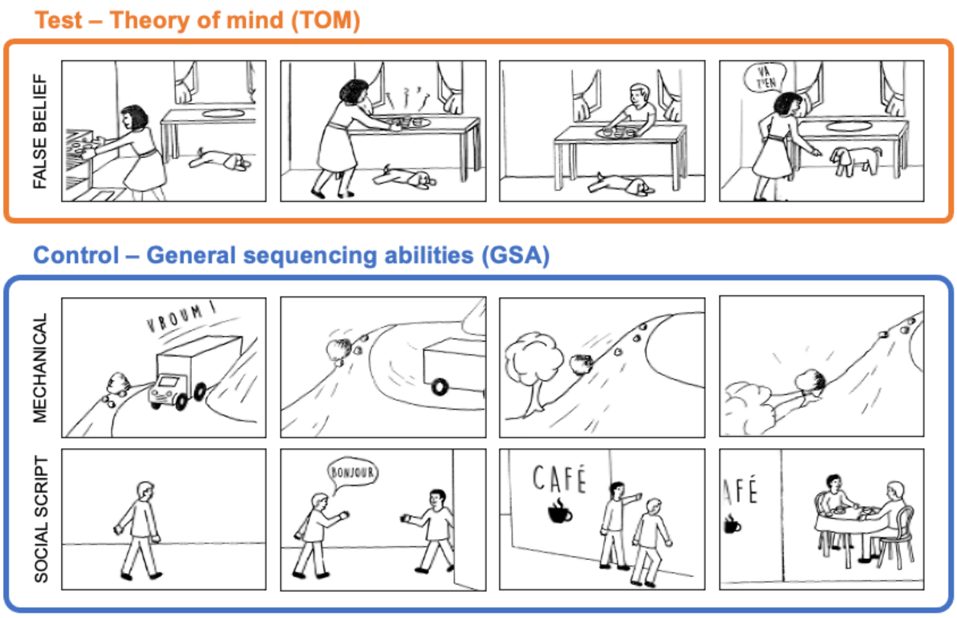

***Figure 1**: PST stories example, involving false beliefs to assess theory of mind (above), or mechanical causalities and social scripts as control (below)*

2019, but also Marocchini, 2023). For example, Senju (2012) proposed that autistic individuals[1] could be able to pass standard false belief tasks when asked to, but that they would struggle to *spontaneously* use TOM to predict others' behavior. This reveals that identifying autistic-neurotypical differences could require subtle experimental designs (e.g., as opposed to simply observing mean accuracy on TOM tasks, see Baltazar et al., 2021). In line with this, Livingston & Happé (2017) argued that autistic individuals might achieve the same level of performance as neurotypical participants in some highly-controlled lab-environments, but that they could rely on atypical processes that might be unusually effortful for them. This explains why some researchers employ response times to better characterize TOM mechanisms (e.g. Behrmann et al., 2006; Livingston et al., 2019). For example, Kaland et al. (2007), who compared autistic to neurotypical adolescents, showed that mental state inferences generally take longer to process than physical state inferences, but that this slowdown was more pronounced in autism.

It is also important to note that most classic TOM tasks used in clinical practice largely rely on language. It is the case of the standard false belief task (Wimmer & Perner, 1983), the Strange Stories test (Happé, 1994), the Faux-pas test (Baron-Cohen et al., 1999) or even the Reading the Mind in the Eyes Test (Baron-Cohen et al., 2001), all of which either use verbally presented stories or rely on a verbal response format. Such tasks are difficult to use in individuals with language disorders because failures can be attributed to either poor TOM or verbal difficulties. This is particularly relevant for autism, which is often found associated with comorbidities, especially language disorders (Levy et al., 2010).

Interestingly, TOM need not be assessed using verbal tasks. For example, Langdon & Coltheart (1999) developed the Picture Sequencing Task (*PST* from now on) precisely to measure TOM without heavily relying on language. In this task, participants are asked to order four picture-cards such that it forms a coherent story. Each picture contains no linguistic material, or very basic 1-word or 2-words utterances (e.g. "Chocolate", "Hi", "Vroom", "Go away"). Critically, these stories can involve characters acting upon false beliefs and, as such, requiring TOM. Two other types of stories are used as controls (see Figure 1): mechanical stories (involving physical causalities) and social-

[1] Following Botha et al. (2021), we use in this paper identity-first language to refer to autistic individuals, judged less offending and stigmatizing than person-first language ("person with autism") by the autistic community (Bury et al., 2020).

scripts (involving everyday social routines). The PST has been used successfully to measure TOM skills in a variety of typical or clinical populations (Langdon & Coltheart, 1999; Payne et al., 2016; Porter et al., 2008; Rajkumar et al., 2008; Van Rheenen & Rossell, 2013).

To date, however, the PST has only been used to compare average TOM skills in groups of patients versus normal controls. The goal of the present study is to evaluate whether a computerized version of the PST (prepared for tablet computers) can be used as a clinical tool in order to assess TOM skills of a single child (compared to normative data). Compared to paper-and-pencil solutions, assessments performed on computer tablets tend to be preferred by children (Piatt et al., 2016) and favor engagement and attention. This is especially the case in autistic children (Lane & Radesky, 2019), who may benefit from such interaction-free and predictable environments. Tablet-based tests also allow for perfectly standardized instruction, execution, feedback and scoring procedures, reducing the risk of experimenter bias or errors, while allowing for reliable group-testing (Bignardi et al., 2021). Moreover, tablets offer an easy and reliable solution to record response times, which would provide relevant complementary information (besides just accuracy)[2]. In this study, we aimed to evaluate whether the tablet-PST was a suitable solution to assess individual children's theory of mind.

EXPERIMENT 1

In order to meet our goal, we begin by testing a large sample of typically developing children with the tablet-PST, and assess its psychometric properties in terms of structural and content validity, and sensitivity to interindividual differences. Provided that such measures are satisfying, this will also allow for the construction of normative data which could be used to derive standardized scores for individual applications.

Methods

Participants

We recruited 248 children in primary school for this experiment (mean age = 8.3 years old, min = 6.0, max = 11.0, 51% female). All participants were native French speakers. Fifty-two participants were in 1st grade, 58 in 2nd grade, 50 in 3rd grade, 42 in 4th grade and 46 in 5th grade. Parental information and consent were collected before testing. All parents declared that their child presented no learning or neurodevelopmental disorder (i.e., autism, attention deficit, or language-related disorder) and no sensory (visual or auditory) or motor disorders that would interfere with their ability to use a tablet. Parents also provided information regarding their socio-economic environment, through (1) the Family Affluence Scale (FAS, Currie et al., 2008), which is a good indicator of family wealth across countries, and (2) their education levels, which were coded on a scale from 0 (no diploma) to 7 (PhD). The FAS and the parental education level were standardized on a scale ranging from 0 to 100 (representing the range of minimum to maximum possible values) and averaged to provide a socio-economic status composite (SES) (mean = 58.4, SD = 16.1). This study was part of a larger project on pragmatic inference skills across development, which received ethical approval from the local IRB (Comité de Protection des Personnes Sud-Est I, ID RCB 2019-A01721-56).

Material

The PST includes four trials of each type (mechanical, social scripts, false beliefs) as well as two training sequences. Each of these will be referred to as an *item*. The minimal verbal content of the original pictures (as depicted in Figure 1) was translated into French. The task was implemented in an application designed for tablet computers. At the beginning of each trial, four pictures were presented in the lower part of the screen in a random order (see Figure 2) while four empty slots were located in the upper

[2] Interestingly, response times measures were included in Langdon & Coltheart (1999)'s original study but were later abandoned, after failing to reveal group difference between high vs. low schizotypal healthy adults. Having seen that response times could reveal meaningful effects in autism, we decided to include this measure to the tablet-PST.

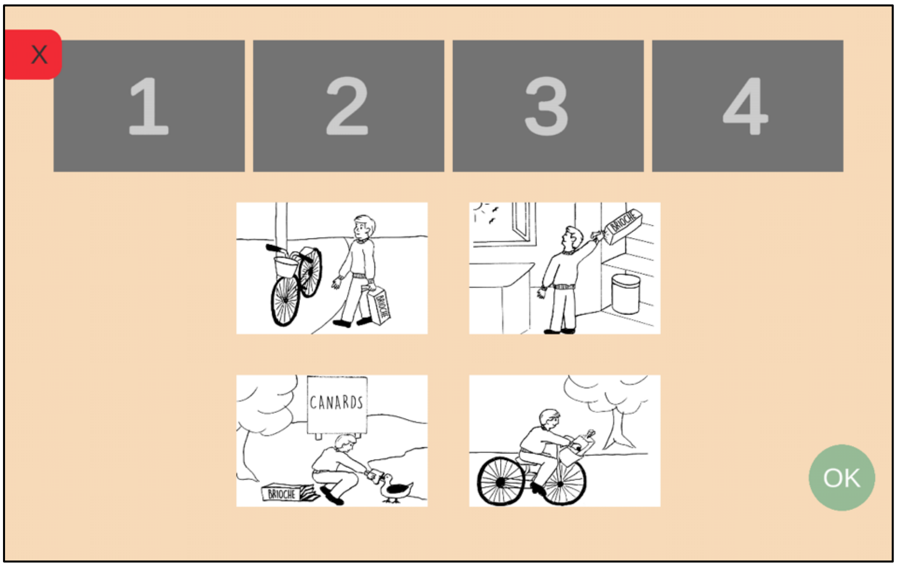

Figure 2*: Task set up for a practice item as it appears on the tablets of the participants*

part of the screen. Children had to select and move the four pictures to the appropriate slots so that the sequence was in order and described a coherent story. They were allowed to modify the sequence as often as they wanted after moving the pictures if they changed their mind. When satisfied, they validated the sequence by pressing *OK* on the bottom right side of the screen. In accordance with the original procedure (Langdon & Coltheart, 1999), accuracy on each item ranged from 0 to 6 based on the appropriate position of each picture (2 points for the first and the last picture, 1 point for the 2nd and the 3rd). Accuracy as well as response time (i.e., time from the beginning of the trial to validation) was recorded by the device at the end of each trial. Trials were pseudo-randomized in a unique trial list. Feedback is automatically provided by the device for the 2 training sequences (a green screen in case of success, a grey screen in case of failure), which children had to succeed to perform the rest of the task (these were repeated as much as necessary).

Procedure

Assessment took place in children's usual classrooms, i.e. in a collective setting, with 15 to 25 children completing the task simultaneously. Each child was provided with a tablet. Automatic audio instructions were included in the task, asking children to move the four pictures to the appropriate slots so that the sequence was in order and described a coherent story, for each sequence. The task began with the two practice items, during which two experimenters remained available and could help children to correctly sequence pictures through verbal guidance. Then the task automatically pursued with the proper items, during which no help was provided to children. Children performed the tasks at their own pace. The task, which lasted about 10 minutes, was performed after other experimental tasks measuring children's pragmatic skills (not reported here) as part of a larger assessment that took about 20 minutes.

Analysis

All analyses were run on R (R Core Team, 2022). These involved three steps. First, we assessed the internal consistency and the factorial structure of the task. Internal consistency was assessed with Cronbach's alpha. The impact of each item on this value was also systematically measured. Factorial structure was assessed with confirmatory factor analysis (CFA) using the *lavaan* package (Rosseel, 2012). Specifically, we compared a unifactorial structure to a theoretical three-dimensions structure.

Second, we used mixed effect models on single trials to analyze accuracy and response times at the group level. In these analyses, we were not interested in the differences among the control items (i.e., between the mechanical and social scripts), but in using them as controls for false-belief items. We thus followed the rationale of Langdon & Coltheart, 1999 (see also Rajkumar et al., 2008 for children) and contrasted TOM items to non-TOM items as a binary variable (*item-type*).[3] The model with accuracy as dependent variable included item-type, age, gender and SES as fixed effects, as well as the interactions of item-type with the latter three. In addition to these factors, the model with response times as dependent variable also included accuracy on each trial and its interaction with item-type as an additional fixed factor. All models included random intercepts for participants and items. Age was included as a continuous predictor (in years with decimals), considering both a linear and a quadratic term (to capture potential non-linear effects). The response time analysis was run on log-transformed values to correct for skewness. Outliers were removed, first roughly at a group level by discarding trials with a response time shorter than 5 seconds or longer than 80 seconds, and secondly at a participant-specific level by excluding trials with a response time greater and lesser than 2.5 SD from each individual's mean RT. Models were fit with the *lme4* package (Bates et al., 2015), sum contrasts were used for all factors, and continuous predictors were centered on the mean. Effects were assessed with Satterthwaite's degrees of freedom from the *lmerTest* package (Kuznetsova et al., 2017), and post-hoc contrasts were computed with the *emmeans* package (Lenth, 2022).

Third, for the sake of preparing the clinical application, we calculated summary measures for each participant and used it to fit linear regressions. These regressions will be used to compare an individual's actual performance to their predicted performance. The advantage of using regressions to calculate discrepancy between a single participant and a reference group is that, instead of considering arbitrarily-defined control sub-groups to provide normative data (for example age-groups or SES-groups), the whole sample size and its variability is exploited to improve the statistical power of standardized scores calculation (Crawford & Garthwaite, 2006). To build these regressions, we use as predictors the factors which were shown to be significant at the group-level (step 2), that is, for instance, we specify age as predictor if and only if it was shown to be linked to performance.

Results

Task validation

Across all items, the overall Cronbach alpha was .75, suggesting acceptable internal consistency. Importantly, removing each item decreased this value (for details, see the supplementary materials, Table S.1.1). The CFA revealed that the three-factor structure (TOM, Mechanical items, Social-scripts) provided a good fit to the data (CFI = .93, TLI = .90, RMSEA = .047), which was significantly better than a unidimensional model ($X^2(3)$ = 17.8, p < .001). The three factors covaried (estimates varied from .32 to .60, all ps < .001).

Group analysis

The linear mixed effect model with accuracy as dependent variable (Table 1) revealed a main effect of item-type, indicating that participants were less accurate on TOM items than the

Table 1*: Estimated parameters of the model explaining accuracy*

Parameter	Coefficient	Standard error	p
(Intercept)	3.14	0.26	*<0.001*
item-type	0.95	0.39	**0.009**
gender	0.05	0.11	0.659
age	22.06	3.09	**<0.001**
age^2	-5.83	3.09	0.059
SES	0.02	0.00	**<0.001**
item-type * gender	0.16	0.14	0.237
item-type * age	-4.95	3.71	0.182
item-type * age^2	8.59	3.70	**0.020**
item-type * SES	0.00	0.00	0.445

[3] To make sure that grouping together social scripts and mechanical did not hide important effects, we also run the analyses contrasting TOM, mechanical and social scripts, and report the results in supplementary materials.

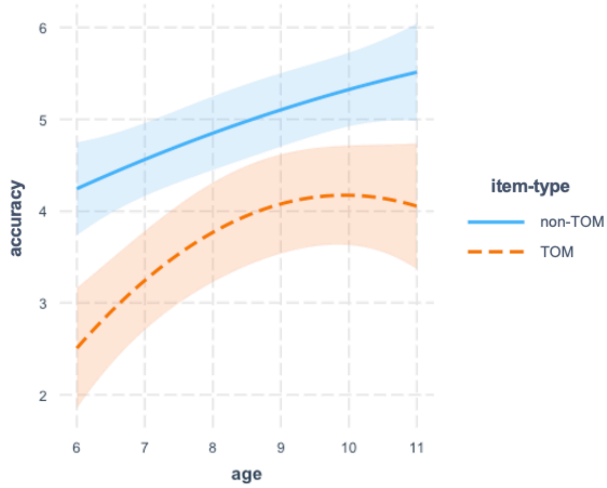

Figure 3*: model's predicted accuracy as a function of age and item-type*

control (non-ToM) items (overall difference ß = 0.96, SE = 0.32, p = 0.01). However, item-type interacted with age: while accuracy on non-TOM items increased linearly with age, accuracy on TOM items reached a plateau at around 9-10 years old (see Figure 3). There was also a main effect of SES, indicating that accuracy generally increased with higher SES. No other main effect or interaction was significant.

The linear mixed model with response times as a dependent variable (see Table 2) revealed a significant interaction between item-type and accuracy (see Figure S1.2 in supplementary materials). While response times did not depend on accuracy for control items (ß = -0.01, SE = 0.01), it increased with accuracy on TOM items (ß = 0.03, SE = 0.01). There was also an interaction between item-type and age. Responses become faster overall with age, reaching a plateau at roughly 9 years of age. TOM items were associated with longer response times when compared to non-TOM items and this difference increased with age.

To make sure that the social script/mechanical story grouping did not hide important effects, we exploratorily reran these two models with 3 item-types instead, which revealed the very same patterns (see the Supplementary materials for detailed models, their outputs and figures).

Individual indices

Summary measures were then calculated for each individual. Specifically, we calculated a TOM-accuracy score by averaging accuracy across TOM items, as well as a General Sequencing Abilities (GSA) score by averaging accuracy on control items (following Rajkumar et al., 2008). Both indices showed satisfying distributions, avoiding floor or ceiling effects across age groups, although the GSA approached ceiling by 5th grade (see Figure S1.5 in the supplementary materials). Log-

Table 2*: Estimated parameters of the model explaining log-transformed response times*

(Intercept)	9.81	0.08	<0.001
item-type	0.07	0.11	0.553
accuracy	**0.01**	**0.00**	**0.004**
SES	0.00	0.00	0.126
age	**-4.32**	**0.99**	**<0.001**
age^2	**2.55**	**0.98**	**0.010**
gender	-0.07	0.04	0.056
item-type * score	**0.03**	**0.01**	**<0.001**
item-type * SES	0.00	0.00	0.175
item-type * age	**2.99**	**0.79**	**<0.001**
item-type * age^2	0.59	0.77	0.446
item-type * gender	-0.01	0.03	0.774

age^2 = quadratic term for age

transformed response times were also averaged for these two item-types (TOM and non-TOM).

Based on the effects observed at the group-level (see above), three linear models were fitted to allow for the calculation of individual discrepancy indices (which will be useful for Experiment 2 below). First, we calculated a model explaining each child's GSA based on their age ($ß = 0.26$, $SE = 0.04$, $p < .001$) and SES ($ß = 0.02$, $SE = 0.004$, $p < .001$). This provides a control index of the extent to which a child is able to sequence pictures in general. Second, we calculated a model explaining each child's *TOM-Accuracy* based on their GSA ($ß = 0.52$, $SE = 0.07$, $p < .001$) and their age with a linear ($ß = 1.94$, $SE = 0.69$, $p < .01$) and a quadratic term ($ß$ -0.0007, $SE = 0.00028$, $p < .05$). This allows one to characterize variations of accuracy on TOM items after regressing out children's age and ability to sequence pictures in general. Third, we calculated a model explaining each child's *TOM-Response times*, based on their accuracy (accounting for speed-accuracy tradeoff, $ß = 0.057$, $SE = 0.01$, $p < .001$) and their own baseline RT on control items ($ß = 0.90$, $SE = 0.05$, $t = 19.0$, $p < .001$). This model allows one to estimate the effort specifically associated with TOM items.

Discussion

In this experiment, we tested a tablet implementation of the PST on 248 typically developing school-aged children. The confirmatory factor analysis provided evidence in support of the structural validity of the tablet-PST, with 3-dimensions opposing items involving mechanical causalities, social scripts and false beliefs. The non-negligeable shared variance between the three factors (also visible in the Cronbach alpha demonstrating internal consistency) confirms the appropriateness of averaging control items together in order to be compared to TOM items, following previous uses of the PST (Langdon & Coltheart, 1999; Rajkumar et al., 2008). More generally, our results are in line with previous studies which used the original task with healthy participants, and which typically observed that TOM items were harder than mechanical and social-scripts items. This is visible with respect to both accuracy (Payne et al., 2016; Porter et al., 2008; Rajkumar et al., 2008; Van Rheenen & Rossell, 2013) and response times (Langdon & Coltheart, 1999). It is interesting to note that children in our sample performed relatively well overall and better than children from a previous study by Rajkumar et al. (2008), who also tested a group of typically developing children (between the ages of 8 and 11). Although it might be argued that the difference comes from the tablet implementation of our task, this is not likely because tablet-devices have provided reliable results compared to paper-and-pencil tests elsewhere (Piatt et al., 2016). Another possibility, which could be investigated in future studies, is that the difference stems from cultural or educational differences between our setting (France) and theirs (Southern India). Critically, however, this difference is orthogonal to the contrast observed between TOM and control items. That is, our results remains in line with the TOM literature, which shows that TOM continues to develop during school age (e.g. O'Hare et al., 2009) and that intelligence in general can account for some, but not all variance on TOM performance (Wellman, 2018). This explains why TOM and the control items are linked but follow slightly different developmental patterns.

Langdon & Coltheart (1999), followed by (Rajkumar et al., 2008), previously used the PST to calculate a TOM composite by subtracting accuracy in control items to accuracy in TOM items. We followed this general approach for our clinical index, except that we determined the coefficient linking control to TOM accuracy on the basis of the group-level data. Moreover, we added age as a supplementary predictor, since we observed that both item-types followed a different developmental pattern. By doing so, our indices are based on theoretical consideration, but are also data-driven, i.e., they are based on the effects which emerged from the reference data. For example, the control index measures how easy it is for a child to sequence pictures in general, was based on age and SES, since both factors played an important role at a group-level, regardless of item-type. Conversely, SES was not included in the TOM-Accuracy index because no interaction was observed between item-type and SES. Including GSA as a predictor of TOM accuracy should thus be enough to account for that variation. For the same reason, gender was not taken into consideration to predict a child's performance. Even though some studies suggested a TOM advantage for girls (Baron-Cohen, 2002), this was not evident in our data, which is consistent with Charman et al. (2002)'s large study who observed this slight advantage, but only in early pre-school development.

EXPERIMENT 2

In Experiment 1, we tested 248 typically developing school-age children with the tablet-PST and observed that it provided outcomes that were consistent with prior findings in the literature. This also constituted normative data against which individual data could be compared. A next important step would be to find support for the task's utility by applying the clinical indices to a sample of children with a condition known to experience difficulties in theory of mind, such as autism. This is what we turn to in the current experiment.

Methods

Participants

We recruited a sample of 23 autistic children (3 girls), mainly through the service for neurodevelopmental disorders at the *Centre Hospitalier Le Vinatier (Lyon, France)*. All the children met the DSM-5 criteria for autism spectrum disorder and received their diagnosis from an experienced child psychiatrist, with no associated intellectual disability, language disorder or attention deficit disorder. Children's assessment results regarding their ASD diagnosis as well as their evaluation of intelligence were extracted from their medical records (see Table S2.1 in the supplementary materials) and confirm the diagnosis. Regarding intellectual abilities, recent measures were not always available (data were not available for three children) and were collected with different instruments, and gathered in different composite, which prevents averaging. However, a careful look at all individual results revealed that all children fall in the range of what is considered normal to normal-superior intelligence (all scores are in the 70-130 range, with most scores in the 100-130 range). All participant attended regular schools. Children were between 6;7 and 10;8 years old (mean = 8.7 years old, SD = 1.36). SES measures were collected as they were in Experiment 1 and were found to be comparable to those of the control group (mean = 55.2, SD = 17.0, $t(25.8) = 1.40$, $p = .17$). Parental information and consent were collected before testing. This experiment was part of the same project than the first experiment, that received ethical approval from the local IRB (Comité de Protection des Personnes Sud-Est I, ID RCB 2019-A01721-56).

Materials

Children were tested with the tablet-PST, as in Experiment 1. To assess convergent validity, children were also tested with another standard 1st and 2nd order false belief task. This task, which relied on verbal stories with picture support and verbal open questions, was taken from the French battery EVALEO 6-15 (Launay et al., 2018). Scores ranged from 0 to 7. To assess divergent validity, parents also fulfilled the French version of the ADHD-RS (Mercier et al., 2016), which assesses inattention and impulsivity symptoms.

Procedure

Children were tested individually in a quiet office of the hospital. Efforts were made to let children progress through the task at their own pace (as in Experiment 1, we avoided to over-monitor what they were doing). Since this study was part of a larger research project, the tablet-PST was proposed together with other cognitive tasks assessing pragmatic skills. Tasks were proposed in an individualized order with as many pauses as needed in order to adapt to each child's level of attention and engagement, but items order within the tablet-PST did not vary from Experiment 1.

Analysis

Our analyses involved three steps. First, we assessed convergent and divergent validity of the tablet-PST. Convergent validity was assessed by calculating the correlation between the mean score on TOM items and the EVALEO false-belief raw score. Divergent validity was assessed by calculating the correlation between the mean score on TOM items and the ADHR-RS questionnaire. Spearman rank correlations were used, given the limited sample size.

Second, we analyzed raw by-trial data at the group level with mixed effect models, by adding the autistic group to the control sample of Experiment 1. As the limited sample size of the ASD group inflated the risk of overfitting and convergence issues, we used the minimally needed models to observe the specific group effects that were of interest. We typically dropped SES which was not implicated in interactions with item-type in Experiment 1, and which did not differ across groups. For both models, we specified group, item-type and age (and their interactions) as fixed effects. We also included random intercepts for items and participants. We then used backward elimination and removed the terms that were not significant from this initial model interaction, in order to reduce model complexity. The model was fitted on all data points for accuracy. For the analysis of response times, we analyzed log-transformed response time data after considering, in line with Langdon & Coltheart (1999), successful responses only (score = 6) in order to reduce complexity in the model. We removed outliers that deviated by more than 2.5 SD in either direction from each group's mean (individually personalized thresholds were not possible due to the exclusion of several items). Models were fit with the *lme4* package (Bates et al., 2015), sum contrasts were used for all factors except group, where TD was used as baseline, and continuous predictors were centered on the mean. Effects were assessed with Satterthwaite's degrees of freedom from the *lmerTest* package (Kuznetsova et al., 2017), and post-hoc contrasts were computed with the *emmeans* package (Lenth, 2022). We specifically reported the group effects and interactions that were of interest and novel compared to Experiment 1.

Third, we computed summary measures for each participant, according to the procedure of Experiment 1, and discrepancy indices based on the normative sample. That is, we used the linear regressions fitted in Experiment 1 to predict each participant's performance and compared it to their actual performance. To qualify this discrepancy, we followed Crawford & Garthwaithe's (2006) methodology, which was shown to be more robust than using the basic model standard error, using the R script developed by Arcara (Montemurro et al., 2022). This discrepancy can be expressed as an estimated percentile rank that ranges from 0 to 100. For comparability, the RT index was reversed, so that a lower estimated percentile rank is always indicative of greater effort or difficulty. We then compared the two groups' standardized scores for the three indices with t-tests and visual examination of the data distributions.

Results

Task validation

As expected, the tablet-PST TOM average score was correlated with the EVALEO false belief score ($\rho = .49$, $p < .05$), indicating convergent validity. Also, as expected, the tablet-PST TOM average score was not correlated with the ADHR-RS total score ($\rho = .16$, $p = .50$), indicating divergent validity.

Group-level results

The linear mixed model explaining accuracy (complete output in supplementary materials) revealed no main effect of group (ß = -0.09, SE = 0.15, $p > .05$) but a group by item-type interaction (ß = 0.45, SE = 0.23, $p < .05$), such that the ASD group performed better than typically developing children with respect to control items (ß = 0.29, SE = 0.27) but worse with respect to TOM items (ß = -0.13, SE = 0.22).

In terms of response times, the model was fitted on 1914 data points (after exclusion of 1.5% outliers, see complete output in the supplementary materials), revealing a main effect of group (ß = 0.24, SE = 0.06, $p < .001$), with ASD participants being slower overall, as well as a three-way group* item-type *age (ß = 0.15, SE = 0.07, $p < .05$) interaction. This interaction revealed that the group difference was not constant over age and item-type (Figure S2.1 in supplementary materials). While TOM items took longer in general than control items, this difference appeared constant over age in the TD group (ß = 0.01, SE = 0.02, $p > .05$) but tended to decrease in the ASD group (ß = -0.10, SE = 0.05, $p = .06$). We thus computed group-differences on this item-type contrast at minimum, mean and maximum age, which revealed that the slowdown associated to the TOM items tend to be larger in the ASD group than in the TD group at 6 ($p = .05$) but not at mean or at maximum ages (ps > .05).

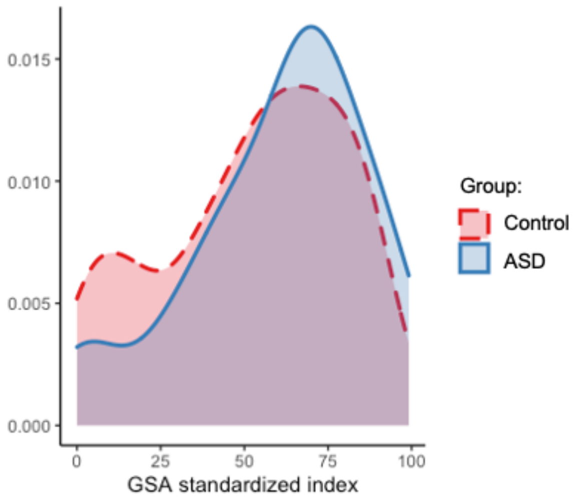

***Figure 4**: GSA standardized index' distribution for the ASD and the control group*

Individual indices

For each child, we estimated their GSA, TOM-Accuracy and TOM-RT percentile ranks, based on the linear regressions fitted in Experiment 1 (note that the results from the control group in this analysis are retrieved from Experiment 1 so they are not new; they are nevertheless displayed for comparison with the autistic group). The Welch two-sample t-tests comparing the groups on those indices failed to reveal significant differences with respect to GSA ($t(26.5) = -1.32$, $p = .20$), TOM-Accuracy ($t(25.7) = 1.52$, $p = .14$) and even TOM-RT which approached significance ($t(27.5) = 1.96$, $p = .06$). However, visual inspection of the distributions suggests interesting effects. On the one hand, it confirms the absence of group-difference for the GSA (see Figure 4). On the other hand, it suggests slightly different distributions for the TOM-Accuracy and TOM-RT indices. Specifically, while the control group appears (expectedly) homogeneously distributed, a pattern emerges for the ASD group. As shown in Figure 5, joint distribution for ASD individuals peaks in the bottom left quarter, where accuracy is poorer and response-times are longer than expected. Although only one fourth of the control group is present in this quarter, half of the ASD sample can be found there ($X^2(1, N = 271) = 5.10$, $p < .05$).

Based on these results (and following Langdon & Coltheart's, [1999] rationale) we computed a TOM-Effort composite, which is an average of

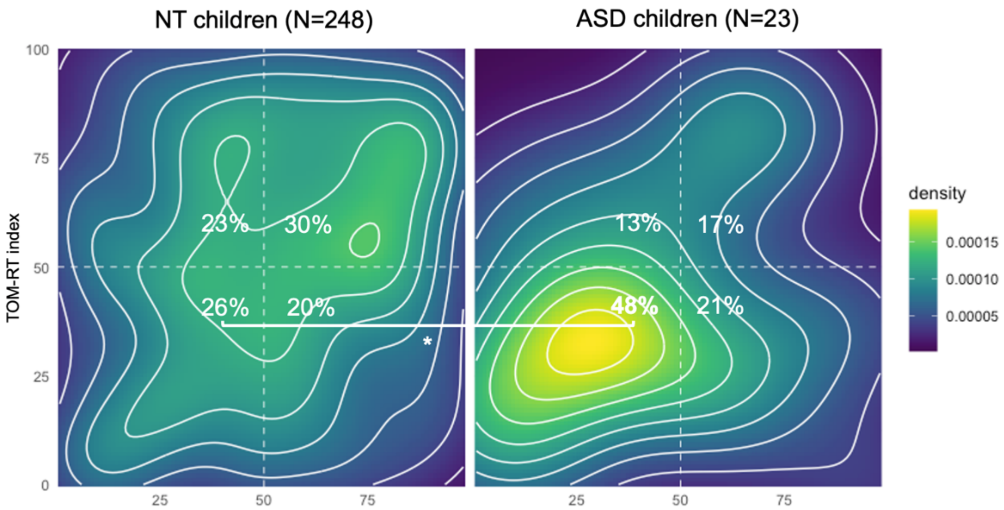

***Figure 5**: Joint distributions of the TOM-Accuracy and the TOM-RT indices for the control group (left panel) and the ASD group (right panel).*

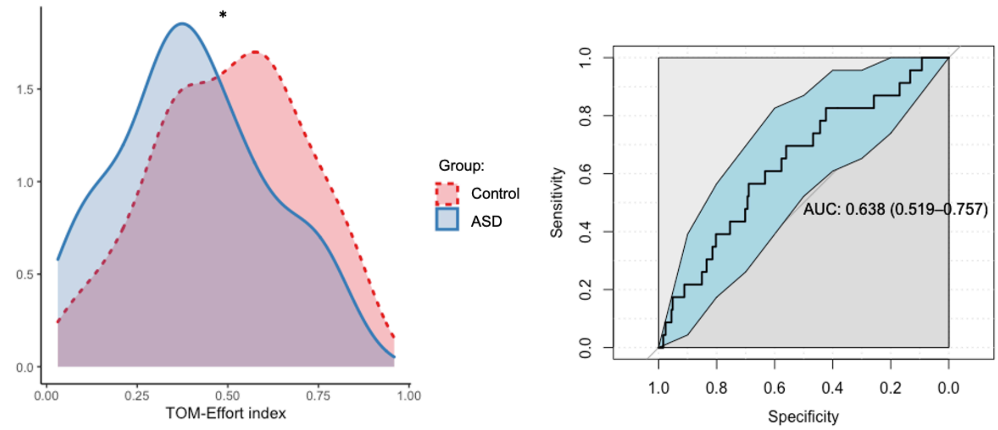

***Figure 6**: Distribution across groups (left panel) and ROC curve (right panel) of the TOM-Effort index*

the TOM-Accuracy and TOM-RT indices. This index revealed a significant between-group difference (t(26.3) = 2.23, p < .05), which is also clear from visual inspection of the distributions (Figure 6, left). To examine whether this composite could discriminate between autistic and typically developing children, we then performed a ROC analysis with this variable (Figure 6, right). This revealed an area under curve (AUC) of .64 (IC = [.52-.76]), which was significantly different from chance but indicated only a moderate accuracy. In appendix, we report on three clinical cases that were extracted from our sample and which illustrate how this task can be used to provide rich clinical information at an individual level.

Discussion

In Experiment 2, the tablet-PST and the clinical indices built from Experiment 1 were applied to a group of autistic children. Before discussing the validity of the tablet-PST in the General Discussion, we first briefly discuss the findings from this experiment at the group level in the context of the literature on autism and TOM. An interesting aspect of our results was an interaction between item-type and group on accuracy, such that ASD children performed at higher rates than those in the typically-developing group on control items and at lower rates on false beliefs items. This dissociation is in line with what Baron-Cohen et al. (1986) originally observed with another picture sequencing task in autistic children. It is also consistent with Binnie & Williams (2003), who showed that when asked to sequence pictures, autistic children tended to prefer physical rather than psychological causalities, compared to typically developing children. This dissociation led to the development of a more general empathizing/systemizing theory of autism (Baron-Cohen, 2009), as a framework to explain the well-described TOM difficulties of autistic individuals. As for response times, we replicated Kaland et al. (2007), who reported slowdowns on TOM items compared to controls, indicating a more pronounced effort in autism, yet only among younger participants in our case. While our sample size calls for caution regarding the interpretation of such subtle effects and for replication in future studies, it is arguably the case that the tablet-PST is not complex or sensitive enough to reveal such differences among older children. The pattern we observe also echoes the results from Zalla et al. (2006) who observed (with a different picture sequencing task) that autistic children were typically slower in sequencing events involving purposive actions taken by characters than physical events, while it was less the case among the typically developing participants.

A novel aspect of our study concerns the use of clinical indices to evaluate TOM impairments at the individual level. Although the combined TOM-Effort index was the only one to show a significant difference between the ASD and the control groups, a visual inspection of the distribution of responses reveals a clear pattern in the autistic group that distinguishes itself from the pattern among typically developing children. That is, TOM items were associated with lower accuracy and slower response times in the autistic group than in typically developing children. It is interesting that the combined TOM-Effort index revealed itself to be the most discriminating, since combining accuracy and response times has also been identified as a promising methods to measure TOM in autistic adults (Livingston et al., 2019). The ROC curve for that measure revealed that it could differentiate the ASD sample from the control group, although with a moderate accuracy (AUC of 64%). Therefore, autistic children should not be expected to systematically fail the tablet-PST (nor should neurotypical children be expected to systematically succeed) and this moderate level accuracy is therefore somewhat expected. Indeed, these data confirm that TOM in general should not be considered *absent* in autism, but rather partial, delayed or simply atypical, the extent of that difference still being under debate (see Livingston et al., 2019; Marocchini, 2023). Hence, discriminating autistic from non-autistic individuals should not be confounded as a validity measure of whether a given task assesses TOM (Chevallier, 2012).

Finally, Experiment 2 also provided evidence for adequate external validity by showing that the tablet-PST provided results that were consistent (in this group of children *without* language disorder) with a standard false belief task; moreover, this association seemed to be specifically explained by TOM and not, say, by the general functioning of children, since no correlation appeared with a control measure of ADHD symptoms, as expected.

GENERAL DISCUSSION

This study aimed to assess the suitability of a tablet adaptation of the PST developed by Langdon & Coltheart (1999), in order to ultimately assess children's TOM clinically and individually. This was achieved by testing a large sample of typically developing children (Experiment 1), providing validation data and allowing for the construction of normalized clinical indices. As a test case, these indices were then applied to a sample of autistic children (Experiment 2).

Overall, our results provided evidence of the validity of the tablet-PST and in several different ways. Structurally, the CFA showed that the task behaved as it was intended. With respect to face and content validity, both the observations on the typically developing group and the autistic sample confirmed the validity of the task, since it behaved consistently to what can be expected from the TOM and the autism literature and, more specifically, with respect to the literature with paper-and-pencil-based presentations of the PST. Both experiments also supported the tablet-PST's discriminatory power, in the general population (with no floor or ceiling effects in the measures) and in a clinical context. We also showed, with respect to our autistic sample, that the tablet-PST provided consistent results with a standard false-belief task (external validity).

However, standard false-belief tasks typically involve verbal content, which is problematic because language difficulties are frequent in developmental populations. In contrast, the tablet-PST is minimally verbal and, as such, more accessible to children with language difficulties. It also offers a suitable solution for research on the TOM-language interface (Bosco et al., 2018). In addition to language, this task also controls for another possible confound observed in most TOM tasks, which is working memory (e.g. Arslan et al., 2017); indeed, with the tablet-PST all the necessary visual information remains displayed while the item is completed. In terms of reliability, compared to the original PST, the inter-judge reliability of this tablet implementation is expected to be almost perfect, since the entire procedure and its scoring are automatized.

As for applicability, the tablet format of the task may help with data collection. Typically developing children frequently reported that carrying out the task was enjoyable, and their teachers described the children as especially focused. This was also the case for autistic children, who remained engaged enough to complete the task, even when parents and clinicians anticipated difficulties. These qualitative observations further support the clinical suitability of the task, and confirmed its usability in group settings (Bignardi et al., 2021).

Our results also stress the importance of using a self-controlled task like the PST and fine-grained indices, which allow one to distinguish participants from two kinds of children who might fail TOM items for different reasons. In one case, it could be due to a general difficulty in picture sequencing, but in another case, it could result from normal sequencing abilities with a specific TOM impairment (both were observed in our clinical sample). Similarly, by exploiting response times, our indices can distinguish between a child who performs in what is considered "typical" fashion for this task from another who performs correctly but exceptionally and effortfully. The indices thus encapsulate clinical reasoning through its algorithms (by integrating control and test items, on the one hand, as well as accuracy and response times, on the other), making it more reliable. This is obviously not to say that such calculation replaces clinical reasoning, but that it provides data that is ultimately finer and more patient-calibrated than what raw performance scores could provide alone. Such data should help drawing a more accurate cognitive and behavioral profile to inform interventions and assess their effects.

We would be remiss if we did not underline three limitations as we consider future research. The first is that, while we had a large sample in Experiment 1, the sample size of our clinical group was limited. This is why Experiment 2 calls for replication and extension, including to other clinical groups, such as children with language disorders or ADHD. Second, in Experiment 2 we compared the clinical sample to the typically developing children who provided the normative data. Further validation would be provided by comparing this group to another, independent group of typically developing children. Third, the cross-sectional methodology we adopted did not allow for test-retest reliability assessment, which could be provided by a multiple testing or longitudinal paradigm.

In sum, after years of research on theory of mind using Langdon & Coltheart's (1999) PST, our study investigates how a tablet-based implementation of this task could be naturally applied to individual clinical assessments. Our data, both in a large typically developing sample and in a group of autistic children, supports the structural, face and content validity of the tablet-PST, as well as its external validity, sensitivity and clinical relevance. The results illustrate how this task can be helpful to assess the profile of children with atypical development, while providing normative data as background for it. This should benefit the assessment of autistic children but also other conditions that might encounter TOM difficulties, such as those with language disorders, attention deficits or psychiatric conditions.

Author note

Authors contributions: NP performed conceptualization, funding acquisition, methodology, investigation, analysis, and wrote the original draft. IN performed supervision and writing – review and editing. MB performed writing – review and editing. JP provided resources and performed supervision, writing – review and editing. We are to thank all the children who participated to this study, as well as Beynes school (Egletons, France) and Saint-Sacrement school (Lyon, France) which welcomed us despite a COVID context, Caroline Morellet and Isabelle Petit who helped with data collection in Experiment 1, Le Vinatier's teams who helped with the recruitment of the clinical group of Experiment 2. This project was funded by the hospital Le Vinatier and the French Ministry of Health (PHRIP-2018-266). The authors have no competing interests, or other interests that might be perceived to influence the results and/or discussion reported in this paper. Data and analytic code to reproduce the analyses are available upon request to the corresponding author.

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Appendix: clinical vignettes

In order to illustrate how the tablet-PST can be used to provide clinical information, richer than a simple standardized score on a classical task, we extracted three prototypical profiles from our sample.

Participant 1 is a 6;7 years old boy. His ASD diagnostic was confirmed by a positive SCQ and a positive ADOS-2. His level of intelligence functioning was also assessed and is strictly normal. In the tablet-PST, he obtained a GSA of 2.8/6, placing him only at the 1^{st} percentile, given his age and SES. This score indicates that the task in itself is not suitable to assess this specific child's skills. Indeed, the ability of that child to sequence picture is compromised, maybe due to his young age, to behavioral difficulties and/or to difficulties in sequencing events and representing time which can be observed in certain individuals with autism (Jurek et al., 2019). If theory of mind must be assessed, it should be done otherwise.

Participant 7 presents a different profile. She is a 7;6 years old girl, whose ASD diagnostic was confirmed by a positive SCQ and a positive ADI-R. Her level of intelligence functioning haven't been formally tested, but no difficulties were suspected by the clinicians who follow her, and her language skills were in the normal to normal-sup range. In the tablet-PST, she got a 4.9/6 GSA (71^{st} percentile) but scored 1.5/6 on average on TOM items (3^{rd} percentile). Contrary to participant 1, participant 7 is thus very able to sequence pictures in general, with a GSA that is even in the normal-superior range. However, specific difficulties arise as soon as TOM is involved, where she scores much lower than expected, given her age and her good sequencing abilities. This is indicative of specific difficulties in theory of mind. No particularity was observed in her response times: she answered TOM items in 29 seconds in average, contra 23 seconds for the control items, which places her at percentile 26.

Finally, participant 6 is a 7;6 years old boy, whose diagnostic was confirmed by a positive ADOS-2 and a positive SCQ. His intelligence was assessed as normal, as well as his language skills. He performed very well on the tablet-PST control, items with a GSA of 5.6/6 (85^{th} percentile), and even better on TOM items with a mean accuracy of 6/6 (94^{th} percentile). This child has no problem in sequencing picture in general and, given his age and his good GSA, he performs even significantly better than expected in TOM items. However, he took 44 seconds in average to answer to items involving false beliefs, contra only 18 seconds on control items: this difference, given his good accuracy, places him only at the 1^{st} percentile. This indicated that participant 6's good accuracy on TOM items is obtained through a clearly unusually important effort, in terms of response times, that is specific to items where TOM was involved. This might reflect atypical processing strategies such as compensatory mechanisms specifically associated with mentalizing.

Supplementary materials

EXPERIMENT 1

Table S1.1: Cronbach alpha value after removing each item (raw alpha = .75)

Item	*Alpha if deleted*	*Item*	*Alpha if deleted*	*Item*	*Alpha if deleted*
mechanical1	.72	**sscript1**	.73	**tom1**	.74
mechanical2	.73	**sscript2**	.73	**tom2**	.73
mechanical3	.72	**sscript3**	.74	**tom3**	.73
mechanical4	.73	**sscript4**	.72	**tom4**	.73

Table S1.2: Estimated parameters of the model explaining accuracy while considering separately mechanical and social-scripts items

Parameter	*Coefficient*	*Standard error*	*p*
(Intercept)	3.30	0.26	<0.001
item-type1	0.14	0.80	0.866
item-type2	1.76	0.80	0.028
gender	0.08	0.11	0.489
age	21.24	3.03	<0.001
age^2	-4.39	3.02	0.146
SES	0.02	0.00	<0.001
item-type1 * gender	-0.23	0.27	0.405
item-type2 * gender	0.55	0.27	0.044
item-type1 * age	12.49	7.40	0.092
item-type2 * age	-22.40	7.40	0.003
item-type1 * age^2	9.25	7.39	0.211
item-type2 * age^2	7.93	7.39	0.283
item-type1 * SES	0.01	0.01	0.180
item-type2 * SES	-0.00	0.01	0.564

Table S1.3: Estimated parameters of the model explaining response times while considering separately mechanical and social-scripts items

Parameter	*Coefficient*	*Standard error*	*p*
(Intercept)	9.77	0.08	<0.001
item-type1	0.09	0.23	0.680
item-type2	-0.22	0.23	0.345
score	0.01	0.00	0.107
SES	0.00	0.00	0.175
age	-4.83	0.98	<0.001
age^2	2.45	0.98	0.012
gender	0.07	0.04	0.058
item-type1 * score	-0.03	0.02	0.080
item-type2 * score	-0.04	0.02	0.027
item-type1 * SES	-0.00	0.00	0.225
item-type2 * SES	-0.00	0.00	0.886
item-type1 * age	-2.72	1.59	0.088
item-type2 * age	-3.27	1.58	0.038
item-type1 * age^2	-1.64	1.54	0.288
item-type2 * age^2	0.45	1.55	0.770
item-type1 * gender	0.07	0.06	0.226
item-type2 * gender	-0.09	0.06	0.137

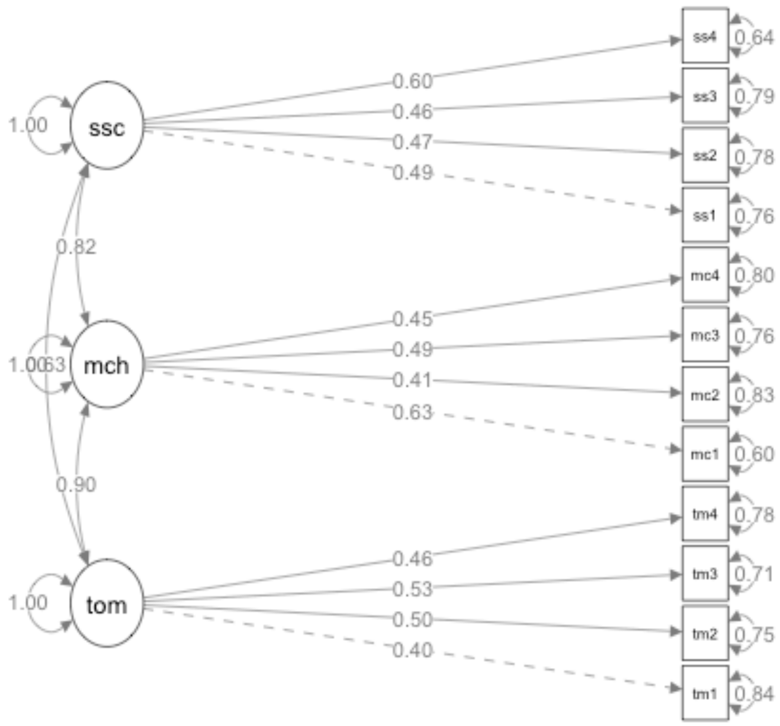

Figure S1.1: Confirmatory Factor Analysis of the tablet-PST

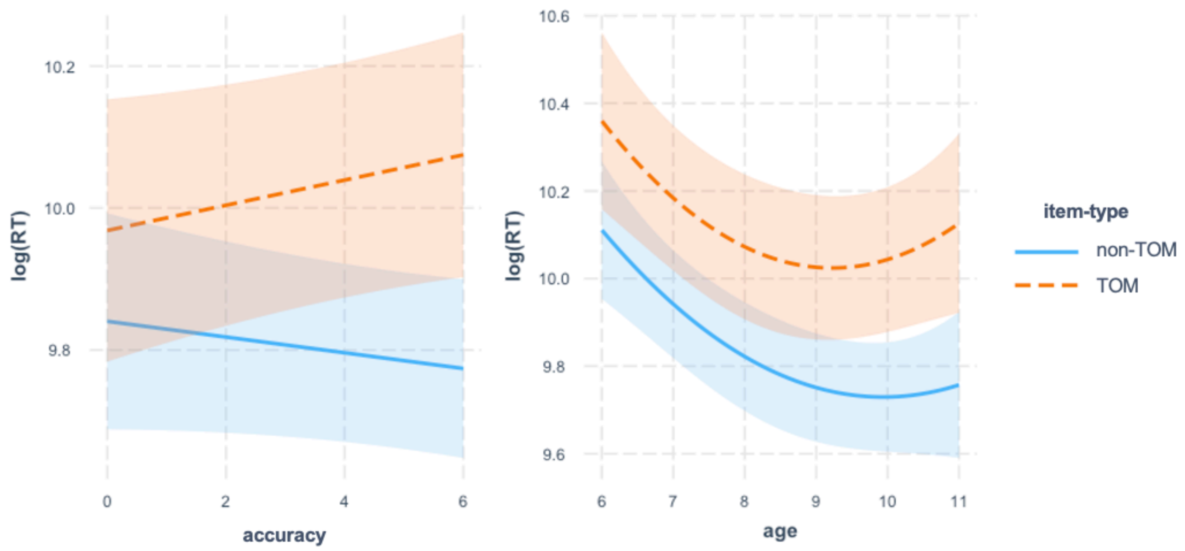

Figure S1.2: model's predicted response times as a function of item-type and accuracy (left) and age (right)

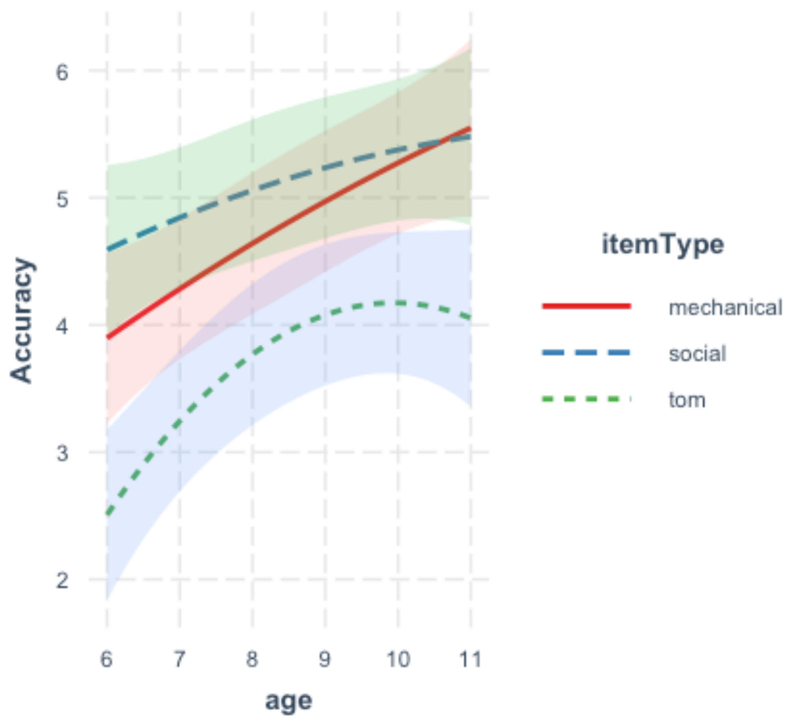

Figure S1.3: model's predicted response times as a function of item-type and accuracy (left) and age (right)

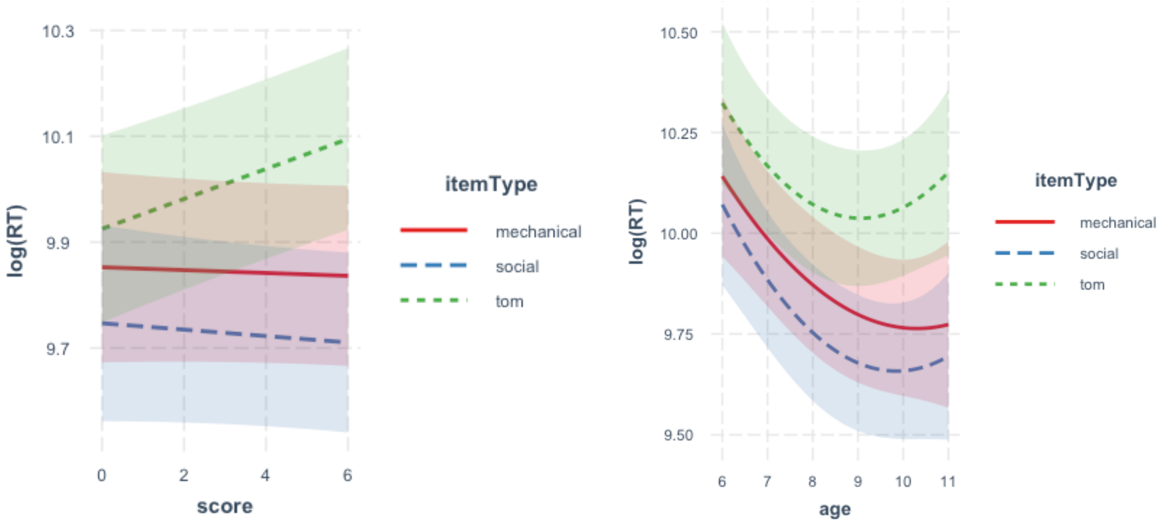

Figure S1.4: model's predicted response times as a function of item-type and accuracy (left) and age (right)

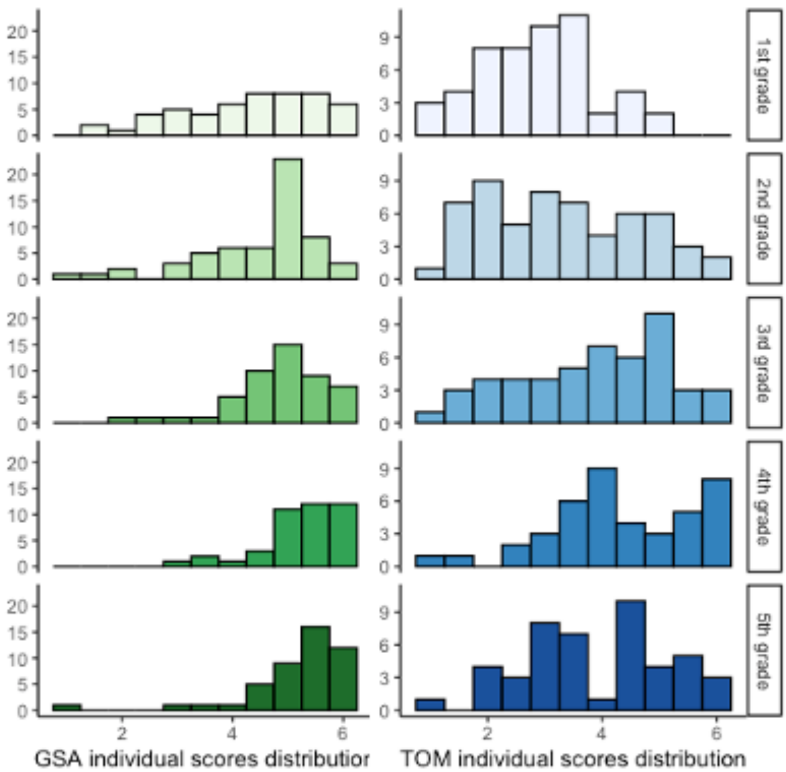

Figure S1.5: Distribution of the two raw composites in the 5 grades-groups

EXPERIMENT 2

Table S2.1: Clinical group detailed description

ID	Age (years; months)	Class	Sex	SES	Autism Assessment	Intelligence assessment*
1	6;7	1st grade	M	80	positive ADOS-2; positive SCQ	FSIQ = 105 (WPPSI-IV)
2	6;7	1st grade	M	17	positive ADOS-2; negative SCQ	*NA*
3	7;1	1st grade	M	50	positive SCQ ; positive ADI-R	VC = 100; VS = 102; FR = 109 (WISC-V)
4	7;2	1st grade	M	20	positive SCQ	FSIQ = 131 (WPPSI-IV)
5	7;2	1st grade	M	65	positive SCQ	FSIQ = 138 (WISC-V)
6	7;7	2nd grade	M	65	positive ADOS-2; positive SCQ	VC = 98; VS = 102; FR = 112 (WISC-V)
7	7;7	2nd grade	F	43	positive SCQ	*NA*
8	7;8	3rd grade	M	35	positive ADOS-2; positive SCQ	VC = 114 ; VS = 117 ; FR = 135 (WPPSI-VI)
9	8;3	2nd grade adapted	F	24	positive ADOS-2; positive SCQ	VC = 92; VS = 89; FR = 97 (WISC-V)
10	8;6	3rd grade	M	49	positive ADOS-2; positive SCQ	VC = 95; VS = 105; FR = 118 (WISC-V)
11	8;6	3rd grade	M	54	positive ADOS-2	FSIQ = 105 (WISC-V)
12	8;9	3rd grade	M	84	positive ADOS-2; positive SCQ	FSIQ = 79 (WPPSI-IV)
13	8;10	4th grade	M	60	positive ADOS-2; negative SCQ	FSIQ = 124 (WPPSI-IV)
14	9;2	3rd grade adapted	M	54	positive ADOS-2	FSIQ = 88 (WNV)
15	9;4	4th grade	M	58	positive ADOS-2; positive SCQ	VC = 98; VS = 102; FR = 112 (WISC-V)
16	9;10	4th grade	M	50	positive ADOS-2; positive SCQ	FSIQ = 92 (WISC-V)
17	10;0	4th grade	F	62	positive ADOS-2; positive SCQ	FSIQ = 100 (WISC-V)
18	10;1	4th grade	M	61	positive ADOS-2; positive SCQ	VC = 118; VS = 138; FR = 118 (WISC-V)
19	10;3	4th grade adapted	M	72	positive SCQ	*NA*
20	10;4	5th grade	M	47	positive ADOS-2; negative SCQ	FSIQ = 112 (WISC-V)
21	10;6	5th grade	M	56	positive ADOS-2; negative SCQ	FSIQ = 115 (WPPSI-IV)
22	10;7	5th grade	M	64	positive SCQ	FSIQ = 127 (WISC-V)
23	10;8	5th grade	M	54	positive ADOS-2	FSIQ = 121 (WISC-V)

* *FSIQ is reported when available. If not, other available summary values are reported. The tests used to provide these measures are detailed.*

Abbreviations used: SCQ = Social Communication Questionnaire; ADOS-2 = Autism Diagnostic Observation Scale 2nd Edition; FSIQ = Full Scale Intelligence Quotient; VC = Verbal Comprehension; VS = Visuo-Spatial; FR = Fluid Reasoning; WPPSI-IV = Wechsler Preschool and

Primary Scale of Intelligence, Fourth Edition; WISC-V = Wechsler Intelligence Scale for Children, Fifth Edition; WNV= Wechsler Nonverbal Scale of Ability

Table S2.2: Estimated parameters of the model explaining accuracy in Experiment 2

Parameter	*Coefficient*	*Standard error*	*p*
(Intercept)	4.28	0.16	<0.001
group [ASD]	-0.09	0.21	0.678
item-type	**1.14**	**0.30**	**<0.001**
age	**0.37**	**0.06**	**<0.001**
group [ASD] * item-type	**0.46**	**0.23**	**0.046**

Table S2.3: Estimated parameters of the model explaining response times in Experiment 2

Parameter	*Coefficient*	*Standard error*	*p*
(Intercept)	9.97	0.06	<0.001
item-type	**-0.36**	**0.11**	**0.001**
group	**0.24**	**0.06**	**<0.001**
age	**-0.11**	**0.02**	**<0.001**
item-type * group	-0.06	0.07	0.446
item-type * age	-0.02	0.02	0.386
group * age	-0.04	0.06	0.512
(item-type * group) * age	**0.15**	**0.07**	**0.042**

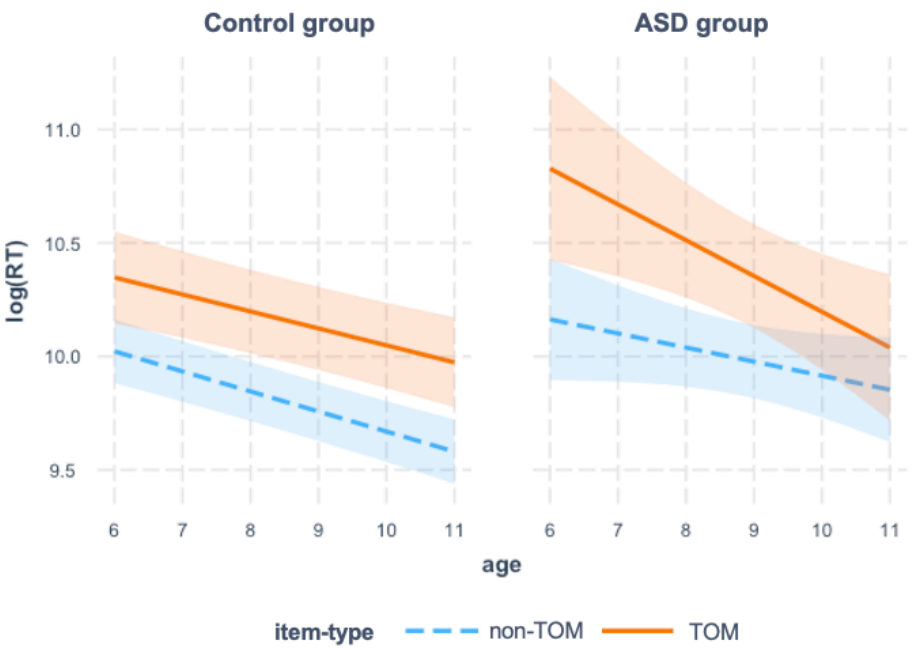

Figure S2.1: Three-way interaction between age, group, and item-type on the response times of succeeded items